I. Problem Proposed based on Stop-and-Wait ARQ

A communication system uses the Stop-and-Wait ARQ protocol to transmit data over a point-to-point link. The system has the following characteristics:

Parameter	Value
Data rate R	2 Mbps
Frame length L	1000 bits
One-way propagation delay T_p	5 ms
Frame error probability P	0.1
ACK transmission time	Negligible
ACK errors	Negligible

Evaluate the followings:

1) Frame transmission time.
2) Normalized propagation delay a.
3) Efficiency without transmission errors.
4) Efficiency with transmission errors.
5) Effective throughput.

Solution:

1)

$$T_t = \frac{L}{R} = \frac{1000}{2 \times 10^6} = 0.5\ ms$$

2)

$$a = \frac{T_p}{T_t} = \frac{5}{0.5} = 10$$

3) ***without errors***

$$\eta_0 = \frac{T_t}{T_t + 2T_p} \rightarrow \frac{1}{1 + 2a} \rightarrow a = 10 \rightarrow 4.76\ \%$$

4) ***with errors***

$$\eta = \frac{1 - P}{1 + 2a} = \frac{1 - 0.1}{1 + 20} = 4.286\ \%$$

5)

$$S = \eta \mathrm{R} = 4.286 \times 2 \times 10^6 = 85.71\ kbps$$

II. Problem Proposed based on Go-Back-N

A satellite communication link employs Go-Back-N ARQ with a sliding transmission window.The system parameters are:

Parameter	Value
Data rate R	5 Mbps
Frame length L	5000 bits
One-way propagation delay T_p	10 ms
Window size N	25 frames
Frame error probability P	0.05
ACK transmission time	Negligible

Assume independent frame errors, a sufficiently large window to keep the link busy, and the standard simplified Go-Back-N efficiency model.

Evaluate the followings:

1. Frame transmission time.
2. Normalized propagation delay.
3. Minimum window size for full utilization without errors.
4. Go-Back-N efficiency without errors.
5. Efficiency with errors.
6. Effective throughput.

Solution:

1.

$$T_t = \frac{L}{R} = \frac{5000}{5 \times 10^6} = 1\ ms$$

2.

$$a = \frac{T_p}{T_t} = \frac{10}{1} = 10$$

3. maintaining continuous transmission without errors, this yield

$$N_{min} \geq 1 + 2a \rightarrow N_{min} \geq 1 + 20 \rightarrow N_{min} = 21$$

As we assumed $N = 25$, the window is adequately large.

4.Efficiency without errors

$$\eta_0 = min\left(1, \frac{N}{1+2a}\right) = min\left(1, \frac{25}{21}\right) = 100\ \%$$

5.Efficiency with errors

$$\eta = \frac{1-P}{1+(N-1)P} \rightarrow \frac{N=25}{P=0.05} \rightarrow \eta = 43.18\ \%$$

6. Effective throughput

$$S = \eta \mathrm{R} = 0.4318 \times 5 = 2.159\ Mbps$$